Tiago Ferrão-Lindqvist

Test & Validation Engineer | Python Data Processing and Reporting

Professional Summary

Test & Validation Engineer transitioning into data engineering, with experience building Python tools for scheduled test-data processing, specification checks and reporting. Developed an internal tool used by four other test engineers and a Power BI dashboard used by approximately 15 engineers. Practical experience with Python, pandas, SQL Server and reusable laboratory data workflows.

Technical Skills

Python: pandas, NumPy, Matplotlib, npTDMS; scripts for TDMS and CSV processing.

SQL and BI: SQL Server, SQL, Power BI; read-only access to the main test database and write access to one IT-created staging table.

Automation: Windows Task Scheduler, LabVIEW.

Personal projects: Flask, SQLite, Git; tutorial-based Docker Compose use.

Engineering: durability and vibration testing, root-cause analysis, MATLAB, Simulink, structural FEA.

Professional Experience

Test & Validation Engineer | Corvane Automotive Components | 2019-present

- Perform durability and vibration testing, root-cause analysis and test-data tooling development.
- Developed Python scripts to process approximately 2 GB of TDMS and CSV data per week from six rigs, flag channels outside test specifications and generate report figures.
- Schedule nightly processing on a laboratory PC using Windows Task Scheduler, writing summary results to an SQL Server staging table created by IT.
- Packaged the scripts as an internal tool now used by four other test engineers for their own tests.
- Built a Power BI dashboard in 2023 showing pass/fail trends and cycles to failure by product generation; used by approximately 15 validation and design engineers.
- Develop MATLAB/Simulink models of seat-adjuster drive loads to select rig load profiles matching customer duty cycles.
- Trained three apprentices in rig safety and test setup during 2022-2024.

Mechanical Design Engineer | Halvik Pumps | 2014-2019

- Designed pump casings and impellers, performed structural FEA and supported performance test-bench work.
- Wrote a LabVIEW acquisition program that reduced test setup time from approximately two hours to 30 minutes.

Personal Project

Bike-sharing trips API | 2025

- Built a small Flask API over a public bike-sharing dataset loaded into SQLite, with three read-only endpoints; runs locally on a laptop.
- Followed a tutorial to run the project with Docker Compose; use Git for personal repositories.

Professional Development

Online data engineering course: six of nine modules completed; no certificate yet.